

AI-POWERED PREDICTIVE MAINTENANCE FOR VEHICLES



MACHINE LEARNING PROJECT

THE PROBLEM

- Unexpected vehicle failures can cause:
 - High repair costs
 - Vehicle downtime
 - Safety risks
 - Disrupted operations
- Traditional maintenance approaches:
 - Reactive: Repair after failure
 - Preventive: Repair according to a fixed schedule



OUR SOLUTION

Vehicle Data → Machine Learning → Failure Prediction → Maintenance Action

The system analyses:

- Engine temperature
- Vibration levels
- Tyre pressure
- Battery status
- RPM
- Usage hours
- Vehicle age
- Maintenance information





BUSINESS VALUE

For Vehicle Owners

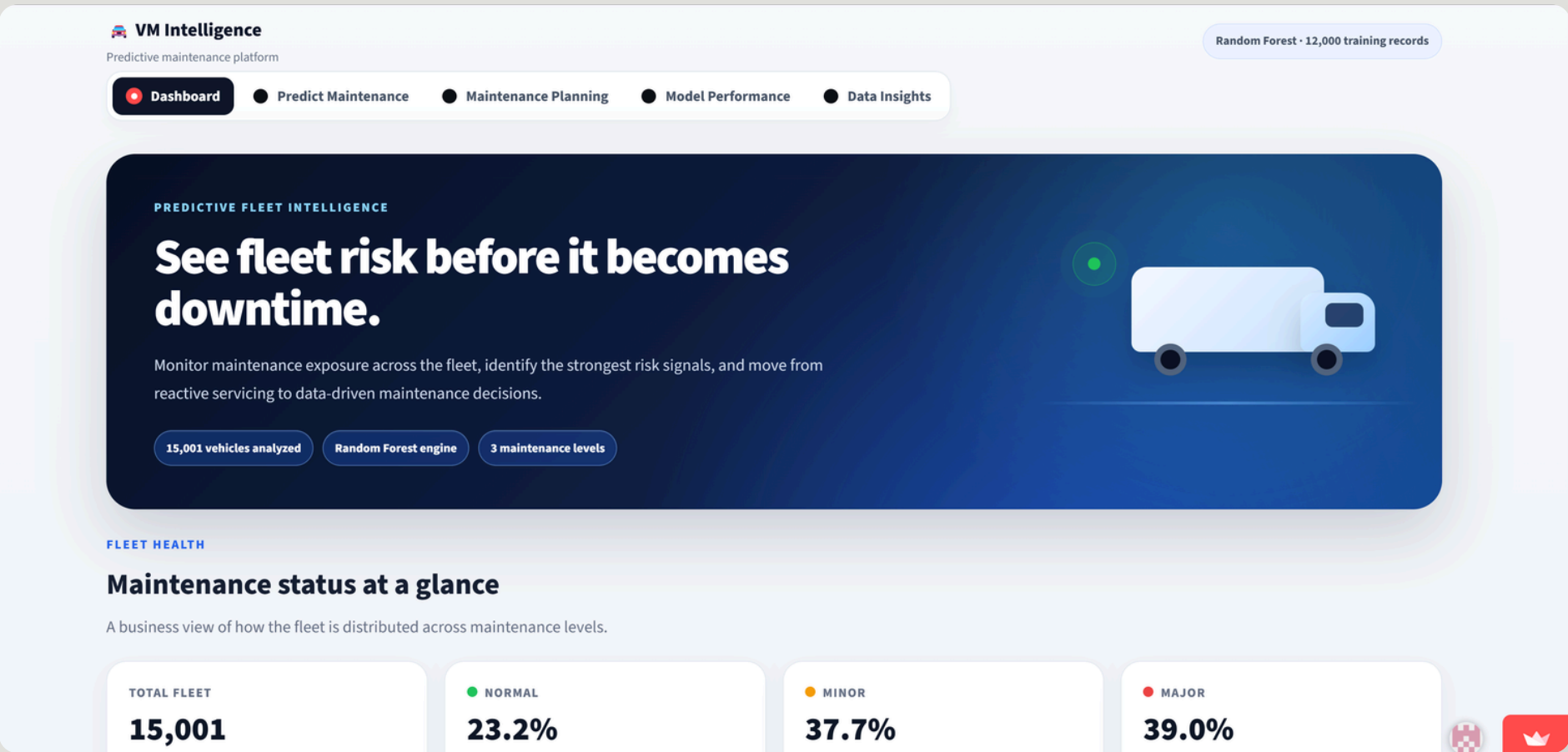
- Reduce unexpected repair costs
- Improve vehicle reliability
- Avoid sudden breakdowns

For Manufacturers

- Identify failure patterns
- Improve vehicle reliability
- Support data-driven decisions

Our Value Proposition:

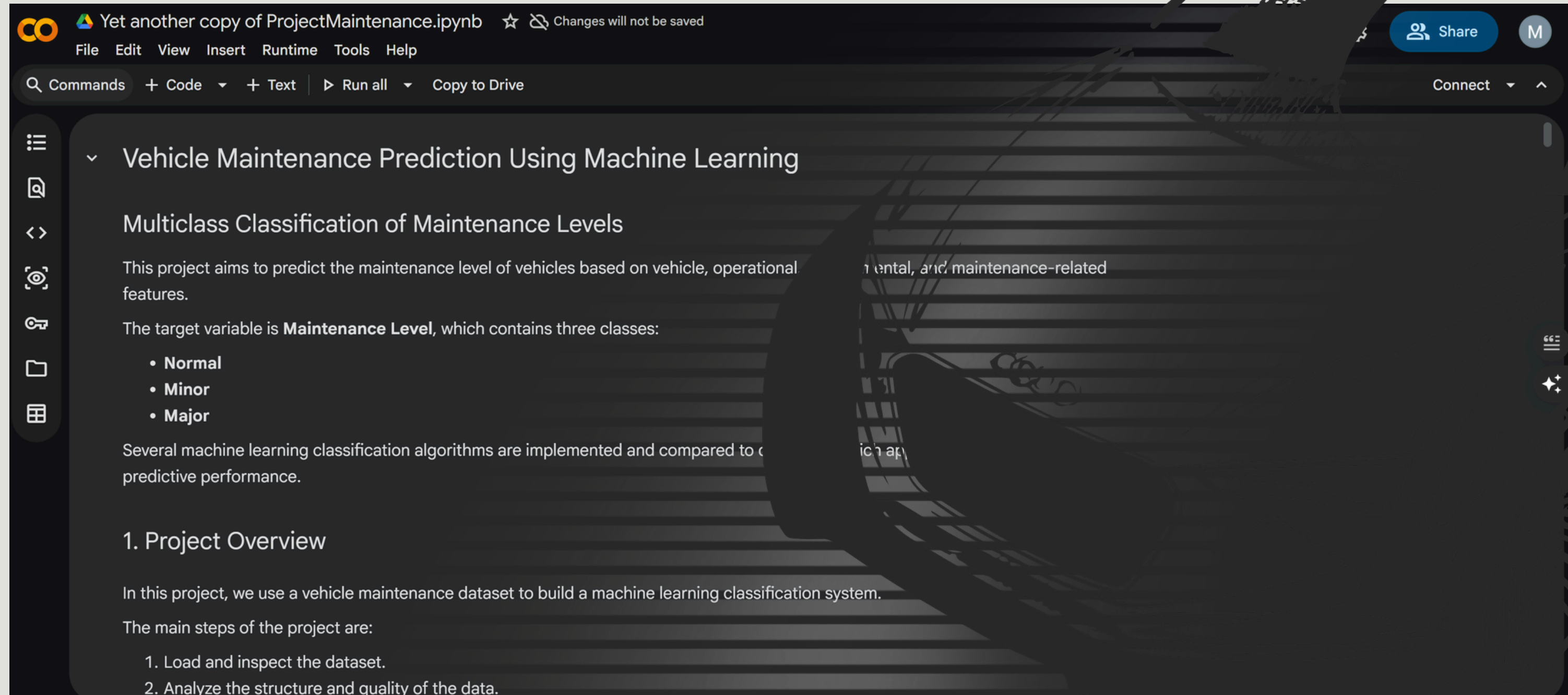
Predict failures earlier. Reduce downtime. Lower maintenance costs.



USER INTERFACE(UI)



THE MODEL



The screenshot shows a Jupyter Notebook interface with a dark theme. The title bar at the top reads 'Yet another copy of ProjectMaintenance.ipynb' and includes a warning 'Changes will not be saved'. The menu bar contains 'File', 'Edit', 'View', 'Insert', 'Runtime', 'Tools', and 'Help'. Below the menu is a toolbar with 'Commands', '+ Code', '+ Text', 'Run all', and 'Copy to Drive'. On the left is a sidebar with icons for file operations. The main content area displays the notebook's title and a detailed description of the project, which is a multiclass classification model for vehicle maintenance levels. The description includes the project's goal, the target variable 'Maintenance Level' with its three classes (Normal, Minor, Major), and a brief overview of the machine learning approach. The first section, '1. Project Overview', begins with an introduction to the dataset and the initial steps of the project.

Yet another copy of ProjectMaintenance.ipynb ☆ ⚠ Changes will not be saved

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all Copy to Drive

Vehicle Maintenance Prediction Using Machine Learning

Multiclass Classification of Maintenance Levels

This project aims to predict the maintenance level of vehicles based on vehicle, operational, environmental, and maintenance-related features.

The target variable is **Maintenance Level**, which contains three classes:

- Normal
- Minor
- Major

Several machine learning classification algorithms are implemented and compared to determine which approach yields the best predictive performance.

1. Project Overview

In this project, we use a vehicle maintenance dataset to build a machine learning classification system.

The main steps of the project are:

1. Load and inspect the dataset.
2. Analyze the structure and quality of the data.

[Take a Look](#)

THANKS

